Mark Garrigan

Lead Full-Stack Software Engineer

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Full-stack software engineer with 25 years building and shipping web products and internal tools across frontend, backend, data, and infrastructure. I start by learning who a project is for and what they need, then plan it, build it, ship it, and keep it maintained and monitored. Since 2024 I've developed agentically by planning each change, guiding AI agents through implementation, and reviewing results before they ship.

TECH STACK

JavaScript	TypeScript	Angular	HTML / CSS	Node.js / Bun	Ruby on Rails	Go	REST APIs	GraphQL
SQL	NoSQL	Docker	Kubernetes	Helm	GitHub Actions	Azure	Stripe	Claude / Copilot

FULL-TIME EXPERIENCE

Optum · Lead Software Engineer

Remote · Apr 2019 – Present

Highlights

- Built and currently maintain the data science platform teams use to set up workspaces and deploy models across cloud and on-prem environments.
- Reduced manual work and support tickets with a self-service engineering portal that automates onboarding, compliance auditing, and credential rotation.
- Built a human-in-the-loop review app so operations staff can check and label LLM decisions, giving data science the feedback it needs to improve model quality.
- Built full-stack internal apps for reporting and for project and resource management.
- Made more than 8,000 commits, pull requests, and code reviews since 2019.

AI-Driven Development

- Moved our team to AI-assisted, spec-driven planning so work is clearly defined before any code is written.
- Ship features with agentic coding, writing the instructions and specs that let AI agents carry out multi-step changes on their own.
- Built Go CLIs: Blackbox (Copilot chat search and recovery) and Plannr (AI agent skills that write specs and keep project tracking current).
- Set guardrails for working with AI agents and documented standard operating procedures regarding the AI software development lifecycle.
- Joined an internal focus group to investigate the feasibility of self-hosted models, benchmarking open-source LLM performance on existing data center GPUs.

Access, Identity, and Secrets

- Built a self-service access request app that gives users one clear view of their access and routes requests to the right approvers.
- Centralized directory and identity lookups into one shared service so each team no longer needs to implement their own.
- Built and currently maintain a centralized configuration and secrets management service adopted by dozens of teams, enforcing secure practices by dynamically injecting secrets at runtime rather than storing them in code.

Platform and DevOps

- Created reusable CI/CD pipelines that give dozens of teams one consistent, hands-off, and repeatable path to production.
- Built remote dev environments so data scientists could run code where their data lives, and simplified how cloud compute is provisioned with one-click deployments.
- Developed Jayz, a custom Go TUI to streamline the testing and consumption of the Azure REST API across internal application environments. It features native Service Principal authentication, instant subscription switching, and dynamic endpoint generation directly from Microsoft Learn documentation.
- Migrated our code to GitHub Enterprise Cloud and managed our organization. Helped protect intellectual property by automating access control, enforcing branch protection policies, and mandating code owners and peer reviews.
- Share on-call duty for production apps, tracking down and fixing issues as they come up.

Security and Tech Debt

- Keep our applications secure and compliant by fixing critical vulnerabilities, rotating credentials, and running yearly disaster recovery exercises.
- Consolidated 13 separately maintained apps and APIs into 2, eliminating 85% of the codebase, which reduced patches, scans, and upgrades and delivered an estimated 200–650 hours of upkeep savings per year.

Architecture and Collaboration

- Architect scalable data and AI solutions while authoring reusable design patterns and technical documentation for engineering teams.
- Mentor and support dozens of engineers and present platform overviews across multiple teams.

FULL-TIME EXPERIENCE CONTINUED

Headway · Senior Full Stack Developer

De Pere, WI · Dec 2015 – Apr 2016

- Engineered robust, full-stack Ruby on Rails applications for startup and enterprise clients, seamlessly translating complex UI/UX designs into scalable, production-ready features across the entire web stack.
- Joined client engineering teams to build applications using Agile methodologies, participating in rapid development cycles and tight feedback loops for immediate stakeholder review.
- Led technical delivery by conducting code reviews, managing project budgets, and consulting directly with clients to map business requirements to reliable technical solutions.

WS Packaging Group · Senior Web Developer

Green Bay, WI · Nov 2008 – Dec 2015

- Led UI and engineering projects for the company's Ruby on Rails products.
- Delivered a zero-downtime digital asset management system handling 18+ million files.
- Built an estimating UI that cut quote generation from days to minutes.
- Led a complete e-commerce overhaul onto a custom platform and built a real-time, Git-based content management system.

SIDE WORK

I love building, learning, and constantly improving my skills, and I spend some of my spare time building things for myself and others.

nurun co. · Founder, Software Engineer & Race Director

De Pere, WI · Apr 2016 – Present

- Founded, direct, and coordinate volunteers for an annual trail and ultra marathon event for up to 500 runners, driving consistent growth primarily through community word-of-mouth.
- Built nurun-admin in 2026 to modernize operations and replace legacy spreadsheets by automating entrant imports, results tracking, household-level logistics, participant communication, and third-party data exports to Bibsee.
- Designed, developed, and iteratively rebuilt the nurun.co race platform to continuously optimize the digital participant experience and streamline event registration.

randmstudio · Co-Founder & Lead Developer

De Pere, WI · May 2005 – Present

- Co-founded a digital agency following four years of freelance web development to build custom software for long-term commercial clients.
- Built custom Ruby on Rails and PHP content management systems tailored to client workflows, or integrated new designs into existing infrastructure.
- Built a local business's first website in 2001 and completely overhauled the platform in 2020 to scale operations through a 580% volume spike in under a year.
- Executed two subsequent full-system overhauls to continually expand features and control, creating a robust infrastructure that has processed over $1M in sales, 20,000 payments, and 7,000 customers via a custom Stripe storefront, order admin, API, driver app, and fulfillment logistics including local delivery.

SIDE PROJECTS

Bibsee · bibsee.com — Live · Sep 2025 – Present

Offline-first race timing. Uses Docker on a Raspberry Pi over local HTTPS, with tablets or phones recording times. Timed the Black River Trail Classic in 2025 and 2026. Solidifying to launch as a product for other race directors.

stored — In development · May 2026 – Present

A self-hosted commerce engine that lets merchants sell without a platform bill before their first sale. It handles products and inventory, Stripe checkouts with live shipping rates, and mailing lists.

boxx — In development · May 2026 – Present

A Go TUI and CLI that runs Docker apps on one Linux host, formed after trying Kamal and ONCE. It gives each app HTTPS through Caddy, an optional MySQL or PostgreSQL database with generated credentials, and zero-downtime deploys.

Plainwork · plainwork.github.io — Live · Dec 2025 – Jan 2026

An experiment in how small a useful app can be. Four macOS menu bar apps, each a single Swift file, plus a Bash notebook CLI, shipped through Homebrew.